

Digital nuclear magnetic resonance spectrometer

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Digital nuclear magnetic resonance spectrometer

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A digital nuclear magnetic resonance (NMR) spectrometer based on a PC is described. The instrument works in a digital manner not only in rf frequency synthesis but also in receiver detection and filtering. A control program and a data processing procedure are developed using the Delphi 5 for Windows '95/'98 operating system. Compared with a traditional spectrometer, the present design is rather simple in its hardware parts. As a consequence, building a NMR spectrometer using the proposed design is simple and low cost. © 2001 American Institute of Physics.

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I. INTRODUCTION

Nuclear magnetic resonance (NMR) spectroscopy has become an important tool for probing the molecular structure and dynamics.¹ To meet the needs of expanding research projects, powerful spectrometers are now commercially available. However in many cases, for a particular purpose one may need only a NMR spectrometer having a subset of the features of a standard commercial one.² Since the application of direct digital synthesis (DDS) to NMR, the design of the spectrometer has been studied extensively.^{2,3} Many papers have focused on the use of DDS for NMR frequency synthesizers.^{4,5} More recently, a digital quadrature detection technique based also on DDS was proposed.⁶ These investigations make the spectrometer design simpler and more flexible.

In this article, we present a homemade NMR spectrometer based on DDS. Its operating frequencies for proton and carbon-13 are 90.0 and 22.7 MHz, respectively. The instrument has three main features: (1) the rf phase and frequency are switched and reset digitally within the frequency source, (2) two-channel quadrature detection is simplified to a single channel, and (3) the pulse programmer is based upon a high-speed 16-bit parallel interface card in a personal computer. The rf phase shifting is done in the frequency source over the entire frequency range. Thus, no extra phase shifting components, delay lines, for example, are required. Since the rf frequency can be reset, it may generate precisely the same rf signal at different times. This characteristic is of great importance when performing digital quadrature detection.⁶ In digital quadrature detection, only one channel of phase sensitive detection is used. Hence, no spectral distortion due to an imbalance between two channels of the conventional detection circuit occurs.

The pulse programmer of our spectrometer is similar to that proposed by Wu *et al.*² Since our spectrometer is greatly simplified, an input/output (I/O) port with 16 bits is enough for normal operation. Moreover, we have developed a pulse

compilation procedure in which users can input and edit their pulse sequence graphically.

The DDS we employed is AD7008, a product of Analog Devices. The clock of the DDS f_{clk} is 50 MHz. Below frequencies of approximately $\frac{1}{3}f_{\text{clk}}$ the output of the DDS is broadband. Since the output of the AD7008 is a sampled signal, its output spectrum obeys the Nyquist sampling theorem. In particular, it contains the fundamental plus aliased signals that appear at $nf_{\text{clk}} \pm f_s$, where $n = 1, 2, \dots$, and f_s is the selected output frequency. Hence, we can expand the output frequency range of the DDS to above 100 MHz without any effort. For instance, we can get a rf near 90 MHz by setting the output of the DDS to about 10 MHz. Therefore, the spectrometer described herein can be easily modified to multiple-frequency operation.

There are two channels with fixed frequencies in our spectrometer. One is for proton and the other for carbon-13 observations. The magnetic field strength of the spectrometer is 2.1 T approximately, corresponding to resonance frequencies of 90.0 and 22.7 MHz for proton and carbon-13, respectively. In the following, we will describe the spectrometer in detail. Finally, some experimental demonstrations will be given.

II. THE SPECTROMETER

A block diagram of the our spectrometer is schematically displayed in Fig. 1. Compared with a conventional spectrometer, simplifications were made mainly in the frequency source, the pulse programmer, and the receiver. The electronic components shown in Fig. 1 may be divided into digital and analog parts. The former includes the PC, the DDS, the I/O, and the analog-to-digital converter (ADC) units, whereas the latter part consists of the rf switch, the rf transmitter, the router, the preamplifier, and the detection components.

There are three AD7008 and 16-bit I/O chips mounted on a printed circuit card that can be inserted into one of the industry standard architecture (ISA) slots in the PC. A block scheme of the DDS board is shown in Fig. 2.

The phase and frequency of the three rf outputs (f_0 , f_1 ,

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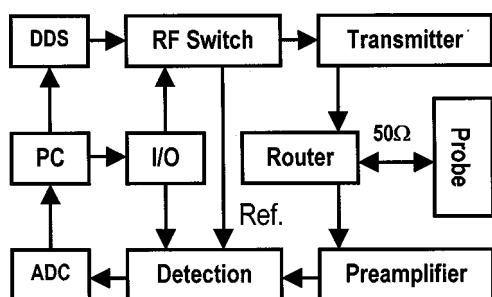


FIG. 1. Block diagram of the spectrometer.

and f_2) with resolution of 10 and 32 bits, respectively, can be set and reset independently by the PC. The first DDS is used for deuterium (^2D) lock, whereas the second and the third DDS are employed for $^1\text{H}/^{19}\text{F}$ and other nuclei. The AD7008 provides a frequency-select pin that permits selection between two frequencies on a per cycle basis. In this article, frequency selection is accomplished through the parallel I/O port. Hence, stable frequency switching is done in less than $0.7 \mu\text{s}$. In addition, the frequencies and phases of the three DDS may be updated within 10 and $4 \mu\text{s}$ directly by the PC through the ISA bus. A 50 MHz OCXO on-board clock is supplied for all three DDS. The controlling logic circuits (the decoder) are built using a piece of field programmable gate arrays (FPGAs) integrated circuit.

A high-speed analog-to-digital converter integrated circuit, ADS804E from Burr-Brown Company, is utilized for data sampling. The ADC unit is designed in such a way that it can be inserted into another ISA slot of the PC. The on-board memory is 2 MB. Since the clock is 20 MHz and the sampling frequency is determined by $f_s = (f_{\text{clk}}/2 + 2N)$, where N is an integer larger than 0, the maximum ADC rate is 5 MHz. Sampling data points and dwell time between adjacent sampling data are transferred through the ISA bus of the PC to the ADC board before the pulse sequence begins. A line from the parallel I/O port is used to trigger this ADC. The controlling logic circuits are built using a piece of FPGA, too.

In the rf switch unit, the rf signal from the DDS is first filtered and then split into the reference and excitation signals. Gating of the rf signals is achieved by a high-speed switching multiplexer (10 ns switch time, AD8180, Analog Devices). A relatively low rf leakage (less than -60 dB) below 100 MHz may be obtained by using this chip. The output voltages (peak to peak) for the reference and the excitation are 1.0 and 2.0 V, respectively.

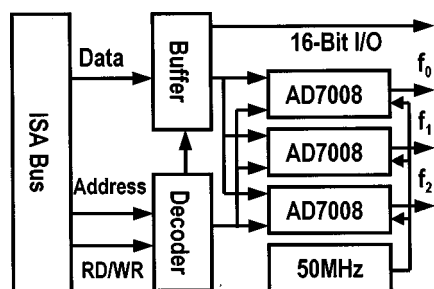


FIG. 2. Block scheme for the DDS card.

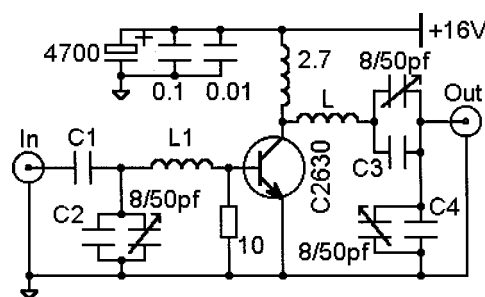


FIG. 3. Schematic of the final rf power amplifier. Component values for the ^1H channel are $C_1=C_2=150 \text{ pF}$, $C_3=51 \text{ pF}$, $L_1=10 \text{ nH}$, and $L_2=150 \text{ nH}$, and for the ^{13}C channel they are $C_1=1.5 \text{ nF}$, $C_2=330 \text{ pF}$, $C_4=300 \text{ pF}$, $L_1=120 \text{ nH}$, and $L_2=850 \text{ nH}$.

The rf pulses from the excitation channel of the rf switch are then fed into the rf transmitter unit, which consists of a rf driver and a rf power amplifier. The rf signal is first amplified up to about 4 W in the driver. The output of the driver is then used to drive the final power amplifier. Figure 3 displays a diagram of the rf power amplifier employed in this spectrometer.

The amplifier is of standard design and operates in class C. A power transistor C2630 (Mitsubishi Semiconductors) is utilized. Because it was possible to build power supplies sufficiently stiff so that no perceptible change in collector voltage occurred during multiple pulse or proton decoupling experiments, at this power level no problem was encountered with "power droop."⁷ The output impedance of the transmitter is adjusted to 50Ω . The final output powers of rf pulses differ between ^1H and ^{13}C channels, although the transmitters are constructed using the same schematic diagrams. In the present spectrometer they are 50 and 200 W, respectively, for ^1H and ^{13}C channels.

The rf pulses from the transmitter are fed into a router, schematically shown in Fig. 4. Diodes D1, D2, D3, and D4 are 1N4148. The values of the capacitors and the impedance in the low-pass π filter may be determined by $C = 0.02(2\pi\nu_0)^{-1}$ and $L = 50(2\pi\nu_0)^{-1}$, where ν_0 is the rf frequency.

Diodes D1–D4 remain on during high-voltage rf pulses. So rf pulses from the transmitter go into the probe without any attenuation because the LC is in parallel resonance. In addition, the preamplifier is protected by diodes (D3 and D4). After rf excitation all four diodes are in the off state.

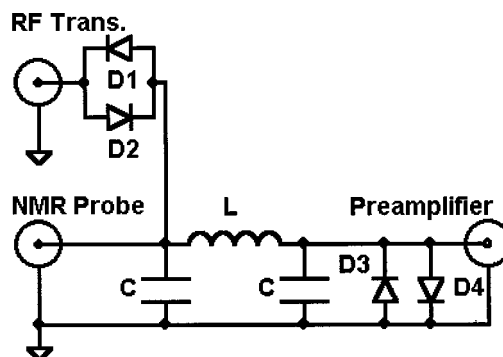


FIG. 4. Circuit diagram of the router. See the text for component values.

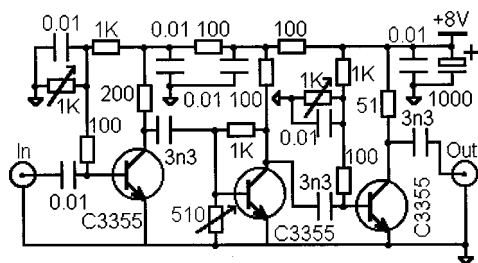


FIG. 5. Circuit diagram of the preamplifier.

The free induction decay (FID) signal from the probe comes into the preamplifier through the low-pass π filter (CLC) without any attenuation. Meanwhile, noise from the transmitter is strongly decreased by diodes D1 and D2.

The receiver consists of a preamplifier, a single channel phase sensitive detector, and an audio amplifier. The preamplifier is built using three low noise transistors and is shown in Fig. 5. A voltage gain of about 60 dB over 5–100 MHz may be obtained for the preamplifier. A noise voltage (peak to peak) of about 200 mV can be found in the output when the preamplifier is connected to the NMR probe (through the router).

The FID signal from the preamplifier is then detected using a quadrant multiplier (AD834, dc to 500 MHz, Analog Devices). Since we employ the digital quadrature detection procedure, only one channel of the phase sensitive detector is used. Finally, the resulting low-frequency FID signal passes through an audio-frequency amplifier (0–60 dB, 0.5–500 kHz) to the ADC input. The main advantage of using an audio amplifier lies in the fact that no dc (very low frequency) drift is present.

Controlling the spectrometer and performing standard functions in NMR experiments are accomplished with the use of a software package. Figure 6 illustrates a functional diagram of the spectrometer software.

A software package was developed using Delphi 5, a program language based on the Windows '95/'98 operating system (OS), to fulfill the functions shown in Fig. 6. The user interface part is employed for operators to set up experimental parameters and to do conventional NMR data processing

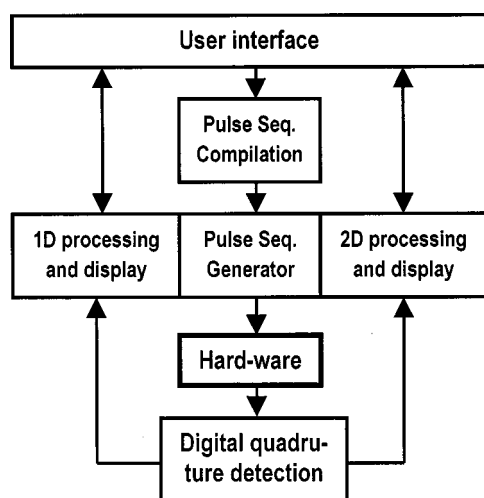


FIG. 6. Simplified representation of the spectrometer software.

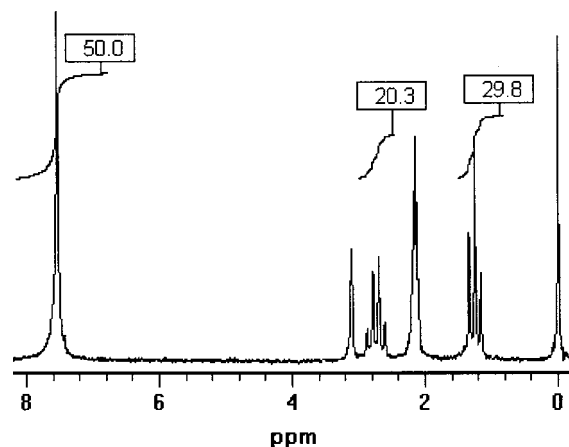


FIG. 7. ^1H -NMR spectrum of 1% ethyl benzene. The integral curves for C_6H_5 , CH_2 , and CH_3 groups are also presented. The experimental conditions are described in the text.

procedures. This part also provides an editor for users to input and edit a pulse sequence graphically. Then the pulse sequence compilation procedure will translate the users input.

The pulse sequence is assumed to be an array of events in which the duration of each event is an integer. Therefore, in the pulse generation procedure, the PC will subtract one from the integer continuously until the integer reaches zero. Each subtraction will take four cycles of the PC clock. So the timing resolution for a pulse sequence running on PC with a CPU clock of 333 MHz is approximately 12 ns. The event, defined as a word type, corresponds to the 16-bit I/O interface. In each event, all 16 bits of the interface are updated, hence DDS frequency selection, rf switching, reference signal gating as well as ADC triggering can be performed. The timing accuracy is an important factor that must be considered, especially when the pulse sequence is running in the Windows '98 environment. An easy way to force the PC to run in real-time mode is to clear any interrupts using the command CLI. A full description of the pulse programmer (pulse sequence compilation and generation) will be given elsewhere.

The principle of the digital quadrature detection (DQD) method⁶ has been proposed recently. The design of the spectrometer may be simplified dramatically by using this scheme, which can be simply described as follows. First, immediately before rf pulsing, reset the DDS and then select the output frequency to f_i (f_i is placed at the center of the spectrum, and this ensures rf excitation efficiency) at once. Second, immediately before detection, switch the output frequency of the DDS from f_i to f_r . The difference between f_i and f_r (Δf) may be several times larger than the spectral width. In this case, the FID signal acquired becomes

$$s(t) = 1/2 \cos[2\pi(\Delta f + \delta) + \Delta\phi] \exp(-t/T_2) \quad (1)$$

where δ is the chemical shift or resonance offset, $\Delta\phi$ represents the phase shift caused by the instrument, and T_2 is the transverse relaxation time. Therefore, the next step is to shift the signal to its normal location. This can be done by multiplying it with a set of quadrature digital sine data $\exp(i\Delta f t)$

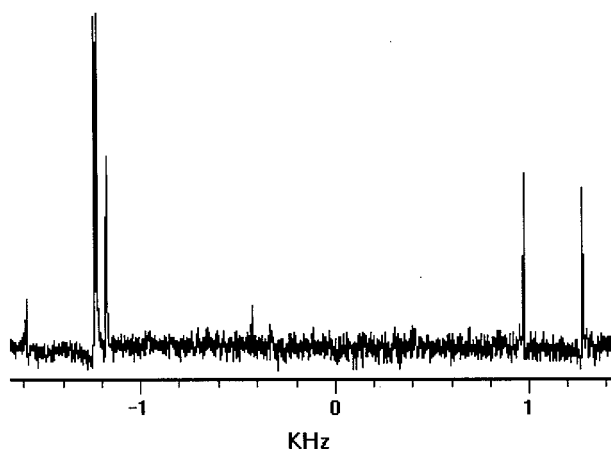


FIG. 8. ^{13}C -NMR spectrum of 90% ethyl benzene obtained with the present spectrometer. The experimental parameters are given in the text.

$+\varphi$). Then the signal is digitally filtered and down sampled according to the desired spectral width, resulting in normal complex FID data. The above data processing procedure can be accomplished during the recovery time interval of each scan.

After digital quadrature detection processing, the FID data are transferred to perform normal one-dimensional (1D) or two-dimensional (2D) data handling.

III. EXPERIMENTAL DEMONSTRATION

We have connected the proposed spectrometer to a 2.1 T permanent magnet which was derived by disassembling an out-of-date continuous wave (cw)-NMR spectrometer (R32, Perkin-Elmer). The rf coil has three turns with diameters of about 6.0 mm circulating on a glass tube with an inner diameter of 5.2 mm. Before the NMR experiment, the probe, a singly tuned resonant in series circuit, is adjusted to $50\ \Omega$ to match the impedance of the router.

Figure 7 is the single pulse sequence ^1H -NMR spectrum obtained with the spectrometer described in this article.

The experimental parameters are as follows: rf frequency $\text{SF}=90\ \text{MHz}$, offset $O_1=11\ 000\ \text{Hz}$, frequency difference between irradiation and detection $\text{DF}=20\ \text{kHz}$, 90° rf pulse width $D_1=2.1\ \mu\text{s}$, spectrometer dead time $D_3=50\ \mu\text{s}$, number of data points for FID acquisition $\text{TD}=2048$ with dwell time $\text{DW}=1\ \text{ms}$, number of scans NS

$=2$, and pulse sequence repetition $D_0=5\ \text{s}$. The actual ADC data points and dwell time are $100\times\text{TD}$ and $\text{DW}/100$, respectively. The oversampled FID is handled in the digital quadrature detection procedure to give a normal FID signal. The rf frequency is derived from setting corresponding DDS outputs to $100\ \text{MHz}$ ($\text{SF}+O_1$)= $9.989\ \text{MHz}$. The FID was zero filling to 16 K before a Fourier transform.

An example of a ^{13}C spectrum obtained with the spectrometer described above is depicted in Fig. 8. The spectrum was acquired using the conventional pulse sequence [nuclear Overhauser effect (NOE)+proton decoupling] and the following parameters, $\text{SF}=22.6\ \text{MHz}$, $O_1=540\ 000\ \text{Hz}$, and $\text{DF}=20\ \text{kHz}$. A 90° rf pulse width of $12.0\ \mu\text{s}$, $\text{TD}=4096$, $\text{DW}=200\ \mu\text{s}$, $\text{NS}=2$, and $D_0=5\ \text{s}$, are used in the data sampling.

So far we have described and demonstrated the NMR spectrometer proposed. The instrument works in a digital manner not only in rf frequency synthesis but also in receiver detection and filtering. Since the spectrometer program and data processing procedures are highly integrated and based upon the Windows '95/98 OS, operation of the spectrometer is simple and easy. Compared with the traditional spectrometer, our design is rather simple in its hardware parts. As a consequence, building a NMR spectrometer using the present design is considerably simple and low cost. The spectrometer presented herein can also be modified to work in real broadbands simply by replacing the AD7008 with a faster AD9852 [300 MHz complementary metal-oxide-semiconductor (CMOS) DDS, Analog Devices] and a broadband rf transmitter.

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